Convolutional Bottleneck Attention for Efficient Hybrid Vision Transformers

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***Abstract*—Hybrid CNN–ViT architectures for edge deployment defer self-attention until the convolutional backbone has reduced spatial extent to 8×8 or 16×16 tokens. The early stages stay pure convolution, where receptive field grows only sub-linearly with depth [1], so full-image context is unreachable until late in the pipeline. The natural alternative — *deep bottleneck attention*: compress aggressively, run attention on the compressed latent, decode back, and place the block at *high* spatial extent — has stayed unoccupied because of two concerns: latent geometry (a class-conditionally clustered latent might be incoherently mixed by attention) and compute cost (a multi-stage non-linear encoder at high spatial extent is too expensive).

The **Convolutional Bottleneck Attention (CBA)** block addresses both. The latent-geometry concern is treated as open: the trained network reaches 81.52% top-1, with three candidate mechanisms catalogued [2] for future verification. The compute concern is resolved by two design choices: convolution over learned projection matrices [3], [4] (preserving the CNN inductive priors that the decoder needs for a spatially-coherent gate), and within the conv family, MobileNetV2-style depthwise-separable convolution [5] (8–9$\times$ cheaper than standard at marginal capacity loss). Together these make a two-stage non-linear encoder around attention tractable at 128$\times$128 for the first time.

CBA-1 sits after stage 1 of MobileViT-S at $128 \times 128 \times 32$, running attention on the $16 \times 16 \times 72$ compressed latent — full-image receptive field at a depth where standard hybrids operate on $\sim$11-pixel local patches. The block integrates via a per-pixel-per-channel sigmoid gate inside an additive residual, $Y = X + \mathrm{DropPath}(\sigma(z) \odot X)$, which preserves the He et al. gradient highway [6] and bounds the multiplier to $(1, 2)$. CBA-MobileViT-S-lite (2.79 M params, 1.114 G MACs), trained from scratch on ImageNet-100 at 256×256 for 50 epochs, reaches **81.52%** top-1 (EMA) — **+4.62 pp** over the matched-protocol MobileViT-XS at $+0.18$ G MACs and **+3.48 pp** over MobileViTv2-0.75 despite a schedule asymmetry that favours the latter.

***Index Terms*—Image classification; vision transformer; mobile and edge deep learning; attention mechanism; depthwise-separable convolution; residual gating; hybrid CNN–transformer architecture; bottleneck attention.**

I. Introduction

A. Where attention sits in a hybrid mobile vision pipeline

Edge-deployed vision models operate under hard constraints (battery, thermal envelope, memory, latency) that data-center models do not. The mobile-vision literature has converged on two responses: efficient CNN families [5], [7] built on depthwise-separable convolutions, and hybrid CNN–ViT architectures [8]–[12] that keep a convolutional backbone for low-level processing and add attention for global context. The hybrid camp is the frame for this work.

Across that hybrid lineage, self-attention is introduced *only after* the backbone has reduced spatial extent to 8$\times$8 or 16$\times$16 — the last 2–3 stages of a 5-stage MobileViT-class backbone. The reason is cost-driven: standard attention scales $O(N^2)$ in token count N, so applying it at the $128 \times 128 = 16{,}384$ token resolution would cost several hundred times more than at $8 \times 8 = 64$ tokens. Every variant takes the same way out: defer attention until the quadratic cost is manageable.

The cost is what's deferred; the consequence is that the *early* stages of every hybrid mobile model remain pure-convolutional. Effective receptive field grows only sub-linearly with depth [1], so full-image context is unreachable until spatial extent has been reduced $\sim$16$\times$ from input. Object-spanning and scene-spanning features that need evidence from distant positions cannot be built at the early stages.

The unexplored alternative. Rather than "wait for the backbone to compress, then run attention on a small token set", *compress aggressively in one block via a learned encoder, run attention on the compressed latent, then decode back*. This places a full-image receptive-field operator at high spatial extent, early in the pipeline, where standard hybrids still operate purely convolutionally. Two structural concerns have kept this design space unoccupied:

1) *Latent geometry.* A classification-trained encoder is widely believed to push same-class inputs together and different-class inputs apart, and attention's softmax-weighted token-averaging is feared to mix this clustering incoherently, producing decoder garbage. Existing precedents [3], [4], [13], [14] sidestep this by keeping the bottleneck linear, compressing only channels (not space), or pretraining the encoder with auxiliary losses.

2) *Compute cost.* A standard 3×3 conv at 128×128×32 costs $\sim$100 M MACs alone; two such stages would dominate the model budget.

The Convolutional Bottleneck Attention (CBA) block addresses both. The latent-geometry concern is treated as genuinely open: the trained model works (81.52% top-1 on ImageNet-100), and Section IV-B catalogues three literature-traceable mechanisms that may explain why. The compute concern is resolved by two ordered design decisions (Section IV-C): **convolution** over learned projection-matrix alternatives (preserving the CNN inductive priors the decoder needs for a spatially-coherent gate); within the conv family, MobileNetV2-style **depthwise-separable convolution** ($\sim$8–9× cheaper than standard at marginal capacity loss). The architectural payoff (Section IV-D): **CBA-1 sits after MobileViT-S's stage 1 at 128×128, with attention computed on the** 16×16×72 **compressed latent** — full-image receptive field at a depth where pure CNN gives only $\sim$11-pixel local features.

B. Contributions

1) **Identification of deep bottleneck-attention as a previously unexplored architectural axis** for hybrid mobile vision transformers, with the two structural concerns above made explicit and existing nearby precedents located on a design-space taxonomy (Figure 1).
2) **An empirical-feasibility position on the latent-geometry concern** based on end-to-end self-consistency: the trained model empirically works (81.52% top-1), and three literature-traceable mechanisms are catalogued as candidate explanations whose verification is future work (Section IV-B).
3) **Resolution of the compute concern in two ordered design decisions**: convolution over learned-projection-matrix alternatives (preserving CNN inductive priors [3], [4]), and MobileNetV2-style depthwise-separable convolutions within the conv family [5] (Section IV-C).
4) **The CBA block** as the concrete instantiation: input → MV2-DSC encoder → MHSA on compressed latent → MV2-DSC decoder → per-pixel-per-channel sigmoid gate inside an additive residual, gradient-safe per He'16 [6] with effective multiplier $1+\alpha \in (1, 2)$.
5) **CBA-MobileViT-S architecture**: 2.79 M parameters, 1.114 G MACs.
6) **Empirical evaluation on ImageNet-100** for 50 epochs at 256×256, against MobileViT-XS (matched-protocol control on the same Blackwell GPU and recipe) and MobileViTv2-0.75 (earlier run on Kaggle T4 under a more favourable schedule, methodologically disclosed). CBA-MobileViT-S-lite: 81.52% **top-1** (EMA), +4.62 pp over MobileViT-XS at +0.18 G MACs and +3.48 pp over MobileViTv2-0.75 despite the schedule asymmetry.

II. Bottleneck-Attention Precedents

Mobile vision backbones. The mobile-CNN lineage [5], [7], [15] established depthwise-separable convolution and inverted-residual blocks as the standard primitives. The hybrid CNN–ViT era [8]–[12] keeps a convolutional backbone for low-level processing and inserts attention only at late stages, after spatial extent has been reduced to 8×8 or 16×16. This deferral is the structural choice the present work re-examines.

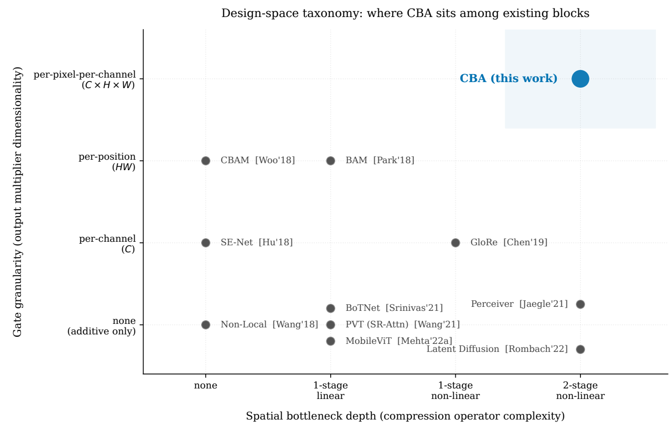

Fig. 1. Design-space taxonomy. x-axis: spatial bottleneck depth. y-axis: gate granularity. CBA (top right) is the unique block combining two-stage non-linear conv compression with per-pixel-per-channel multiplicative gating, trained end-to-end by classification loss alone.

Attention efficiency. Two parallel lines reduce attention's cost. The first manipulates the token *count* — DynamicViT [16], EViT [17], ToMe [18] — by routing on signals derived from a full attention pass. The second — the lineage CBA extends — manipulates the *space* over which attention is computed by inserting a bottleneck before the attention sublayer. The precedents most relevant to CBA:

- **PVT** [4]: Spatial-Reduction-Attention via a single strided 1×1 conv on K, V before attention. Linear (no nonlinearity), one stage.
- **Linformer** [3]: K, V projected through a learned matrix $E \in \mathbb{R}^{N \times k}$ to a fixed-length sequence. Linear and *token-dimension* compression — no spatial inductive bias whatsoever.
- **BoTNet** [13]: replaces the 3×3 conv in the last three ResNet bottleneck blocks with MHSA — but at *full* spatial resolution; only *channel* compression.
- **Perceiver** [14]: small set of learnable latents updated by cross-attention to the input. Standalone architecture, not an in-CNN-block residual gate.
- **BAM** [19]: bottleneck attention modules at network bottlenecks, but uses MLP-with-ReLU + dilated-conv rather than self-attention.

The taxonomy is summarised by the design-space scatter in Figure 1: spatial bottleneck depth × gate granularity. **No existing block occupies the position** (two-stage non-linear conv bottleneck) × (multi-head self-attention on compressed latent) × (placed at high spatial extent) × (trained end-to-end by classification loss alone). CBA enters this gap.

III. Preliminary Investigation: Pre-Attention Gating

Before settling on the CBA design, we tried a different idea: if a cheap closed-form score could pick which tokens the transformer actually needs, we could throw the rest away pre-attention and recover compute. The result was negative. This section documents what we tried, what we measured, and what we learned — because the failure shape is what motivated the architectural pivot.

A. Original hypothesis

The convolutional branch of a hybrid model produces, before any attention runs, a feature map $X \in \mathbb{R}^{B\times C\times H\times W}$ that already encodes spatial saliency. We hypothesised that a cheap, closed-form scoring function $s\colon \mathbb{R}^{C} \to \mathbb{R}$ applied per spatial position could rank tokens by attention-relevance, letting us keep only the top $K = \lfloor N\rho \rfloor$ at keep ratio ρ and bypass the rest through the existing residual. Token reduction in pure ViTs [16]–[18] achieves 30–50% sparsity at <1 pp loss but always *post-attention*; a free CNN-derived *pre-attention* signal would be strictly cheaper.

B. Experimental design

We instrumented MobileViT-S [8] with five non-learned scoring functions on the per-position channel vector $x_{i,j} \in \mathbb{R}^{C}$ — normalised Shannon entropy, L^1 magnitude, L^2 norm, cross-channel variance, and max-channel activation — selecting the top-K at keep ratios $\rho \in \{0.3, 0.5, 0.7, 1.0\}$. Bypassed tokens passed through MobileViT-S's existing `conv_fusion` residual; selected tokens flowed through the unmodified transformer block.

C. Results

The headline result is for the entropy gate at ρ=0.5 on ImageNet-100. Baseline (no gate): 91.64% top-1, 308 img/s on a T4. Zero-shot with the gate: 1.12% (chance). After 5 epochs of full-model fine-tuning: 83.64% at 392 img/s — -8 pp accuracy for $+27\%$ throughput. The other four scoring functions clustered within run-to-run noise of entropy. The trade-off is dominated by the trivial alternative of running MobileViT-XS at higher throughput with no gating logic.

D. Why the closed-form approach failed

A scalar reduction $s(x_{i,j})$ of a C-dimensional channel vector at one spatial position is monotonic in some statistic of activation magnitude. The property "this token, in this layer, will benefit from full self-attention" is not such a function: it depends on which other tokens populate the sequence, which queries the learned W_Q, W_K, W_V will form, and which downstream task is being optimised. Per-position channel statistics encode none of these.

Three avenues remain to recover a workable pre-attention gate: (a) learn the gate as a sparsity-regularised scoring network [16] — forfeits the "free" signal that motivated the hypothesis; (b) gate post-attention [17], [18] — the $O(N^2)$ cost has already been paid; (c) redesign the attention block so gating is intrinsic to its operation rather than a pre-filter. The CBA block develops avenue (c).

IV. Design Hypothesis and Evolution

A. Deep bottleneck attention as an unexplored axis

The lesson from Section III is that gating must be *intrinsic* to attention, not a pre-filter applied on top. The shape of that redesign is the unfilled position in Figure 1: a two-stage non-linear conv bottleneck encoder, self-attention on the compressed latent, placed early in the pipeline, and trained end-to-end by classification loss alone. Two structural concerns explain why the position has stayed unfilled — latent geometry and compute — and the next two subsections address them in order.

B. Concern O1: Latent geometry under task supervision

A classification-trained encoder is expected to cluster same-class inputs together and push different-class ones apart, and attention's token-averaging is feared to mix this clustering incoherently into decoder garbage. This is the structural reason behind the otherwise curious design choices in nearby precedents: PVT keeps the bottleneck linear; BoTNet compresses only channels; Perceiver regularises the encoder with auxiliary losses.

We treat O1 as an open empirical question. What we can claim is that the architecture trained end-to-end by classification loss *works*: 81.52% top-1 on ImageNet-100, monotonic training with no rank-collapse signatures, and visibly object-localised gate maps (mean $\bar{\alpha} \in [0.10, 0.18]$ with maxima approaching 1.0; Figure 2). The encoder, attention, decoder, and gate are jointly optimised by a single classification loss with no auxiliary objective, so the encoder's latent is, by construction, whatever makes the downstream attention output useful for classification. This is a feasibility claim, not a mechanistic explanation of why such a fixed point exists.

Three candidate mechanisms in the literature could underlie that fixed point. First, *token identity*: attention here operates on the N=256 tokens of a single forward pass — all derived from the same image — so its mixing is within-image, never across-class clusters. Class-conditional clustering is a dataset-level property, not a per-image one. Second, the *Information Bottleneck* [2]: deep VIB [2] reports class-conditional clustering at high β with within-class variation preserved as task-relevant signal, and if CBA's deterministic encoder reaches a comparable point, attention operates in the preserved within-class subspace. (IB theory is stochastic, so this remains an intuition.) Third, *working precedents*: PVT [4] and BoTNet [13] both run self-attention on task-supervised compressed latents and train cleanly, and Linformer's Theorem 1 [3] bounds attention's effective rank to $\Theta(\log n)$ — attending on a low-rank latent loses essentially nothing the original tokens could express. None of these is CBA, but each demonstrates the operator class is feasible.

Empirically distinguishing these mechanisms (per-image vs. across-image latent geometry; within-class vs. between-class cosine and linear probes; rank profile across depth) is future work (Section VIII-D); the architectural contribution does not depend on it.

C. Concern O2: Compute cost, resolved by depthwise-separable convolutions

A standard 3×3 conv at CBA-1's 128×128×32 input with 32→64 channels costs ∼302 M MACs; two such stages would cost ∼600 M — half the entire MAC budget of a MobileViT-XS-class model. This is the arithmetic that has steered every nearby precedent to a less expressive operator. We resolve it with two ordered choices: (D1) convolution over learned-projection-matrix alternatives, and (D2) within the conv family, MobileNetV2-style depthwise-separable convolution.

1) D1. Convolution over learned-projection-matrix alternatives: The cheaper precedents — PVT's strided 1×1 conv [4] and Linformer's learned token-axis projection [3] — discard structural priors that vision needs. Linformer's projection in particular has *no spatial inductive bias*: nothing in its learnable matrix encodes that adjacent tokens should mix more strongly than distant ones, that translation should be equivariant, or that output tokens retain spatial identity. A 3×3 conv encodes all three structurally; a free learned matrix would have to discover them from data, unaffordable in our small-data regime (∼126 K IN-100 images, no IN-1K pretraining, 50 epochs).

There is a second reason for conv: spatial identity in the latent. Each token at output position (i', j') corresponds to a receptive-field region of the input. This is what the decoder maps back to a spatially-coherent gate, and what makes Section IV-D's receptive-field claim well-defined. With a Linformer-style projection, the k output tokens are abstract learned combinations with no spatial identity — there is nothing for the decoder to "decode back to".

2) D2. Depthwise-separable convolution over standard convolution: A standard $D_k \times D_k \times C_{\text{in}} \times C_{\text{out}}$ kernel factors as a depthwise spatial filter followed by a pointwise mixer:

$$K(p, q, m, n) \approx K_{\text{DW}}(p, q, m) \cdot W(m, n), \tag{1}$$

with cost ratio $\text{MACs}_{\text{DSC}}/\text{MACs}_{\text{std}} = 1/C_{\text{out}} + 1/D_k^2 \approx 1/9$ at D_k=3 — about 8–9× cheaper per layer with no classification-capacity loss at scale [7], [15].

The MV2 inverted-residual [5] refines this for the bottleneck setting: PW expand → DW at expanded width → linear PW project. For CBA-1's 128×128×32 → 16×16×72 encoding, the two-stage MV2-DSC encoder costs ∼36 M MACs versus ∼600 M for a standard-conv equivalent — the difference between tractable and impossible at high spatial extent. The decoder mirrors the encoder with bilinear upsampling inserted between PW-expand and DW; bilinear is artifact-free and cheaper than transposed convolution.

D. Architectural consequence: full receptive field at high spatial extent

With O1 and O2 addressed, the headline architectural payoff follows: **attention can be placed at high spatial extent, granting full-image receptive field at a stage where standard hybrids still operate purely convolutionally.**

CBA-1 sits after stage 1 of MobileViT-S at $128 \times 128 \times 32$, compresses to a $16 \times 16 \times 72$ latent (s_1=4, s_2=2), runs L=2 MHSA layers on the resulting 256-token sequence, and decodes back to $128 \times 128 \times 32$. Attention's $O(N^2)$ cost is paid at N=256 (∼65 K pairwise interactions) — never at the N=16,384 a direct 128×128 attention would require.

In a stacked CNN with D_k=3 kernels and stride-2 down-samples every other stage, the effective receptive field grows sub-linearly with depth [1]. By the end of stage 1 of MobileViT-S, each output position has only ∼11-pixel local access — roughly 4% of input extent. CBA-1 replaces this with full-image receptive field in a single block: every position in the 16×16 latent attends to every other, and the decoder maps the result back to 128×128, so every output position has received contributions from every input position via the attention pathway. CBA-2/3/4 follow the same template at progressively reduced spatial extent $(32^2, 16^2, 8^2)$; CBA-1 is the load-bearing instance.

Figure 2 shows the qualitative footprint: CBA-1's gate maps are the most object-localised of the four blocks, consistent with the spatial heavy lifting happening at the early stage.

E. Supporting choice: positional encoding is omitted

CBA omits all explicit positional encoding. The DSC encoder's two 3×3 depthwise convolutions with zero-padding implicitly encode position via the PEG mechanism [20], and the literature consistently shows that explicit PE adds nothing when convolutions surround attention [8].

F. Supporting choice: FFN is retained

The FFN is retained at modest expansion factor f=2 (vs. ViT/DeiT's f=4). MetaFormer [21] shows the channel MLP is the load-bearing component: IdentityFormer-M48 with no token mixing reaches 80.4% IN-1K. The FFN also raises the per-layer Lipschitz constant, slowing rank collapse in deep stacks of attention layers.

G. Supporting choice: residual integration via Hadamard inside additive shortcut

We integrate the post-attention output z with the input X as $Y = X + \alpha(X) \odot X = (1 + \alpha(X)) \odot X$ with $\alpha = \sigma(z) \in (0, 1)^{B \times C \times H \times W}$. Three properties justify the choice.

Proposition 1 (Hadamard-vs-additive rank growth)**.** *For* $A, B \in \mathbb{R}^{m \times n}$*,* $\text{rank}(A+B) \leq \text{rank}(A)+\text{rank}(B)$ *(linear), whereas* $\text{rank}(A \odot B) \leq \text{rank}(A)\cdot\text{rank}(B)$ *(Schur 1911 [22], [23]; quadratic).*

CBA's residual $\alpha \odot X$ thus has rank up to $\text{rank}(\alpha)\cdot\text{rank}(X)$ — quadratic, strictly dominating the rank-1 (per-channel) gate of SE-Net.

Proposition 2 (Gradient highway preservation)**.** *For* $Y = X + \alpha(X) \odot X$ *with* $\alpha \in (0, 1)^d$*,* $\partial Y/\partial X = \text{diag}(\mathbf{1} + \alpha) + \text{diag}(X)\nabla_X \alpha$*, so* $\sigma_{\min}(\partial Y/\partial X) \geq 1 - \|X\|_\infty \|\nabla_X \alpha\|_2$ *by Weyl's inequality [23]; under gate saturation* ($\nabla_X \alpha \to 0$)*,* $\sigma_{\min} \geq 1$ *unconditionally.*

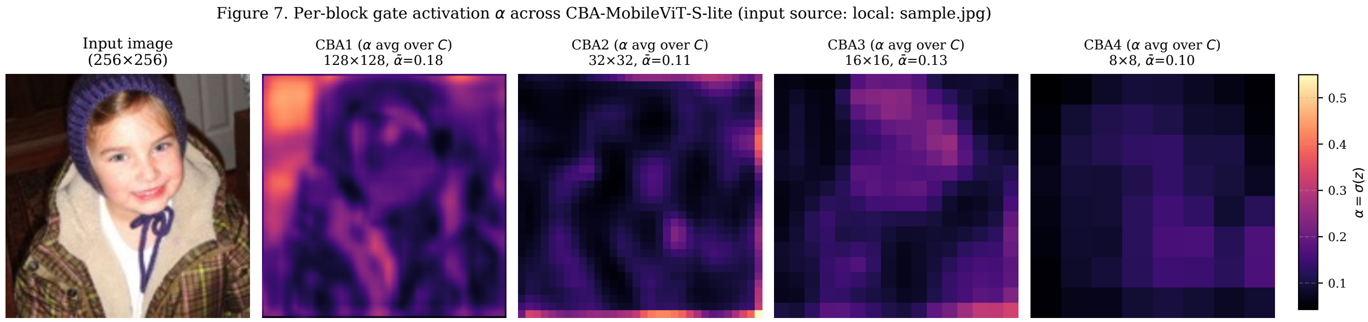

Fig. 2. Per-block gate activation $\alpha = \sigma(z)$ from the trained CBA-MobileViT-S-lite checkpoint on an ImageNet-100 validation image. CBA-1 (at 128×128 spatial extent) shows the most pronounced object-localised gate structure of the four blocks, consistent with the early receptive-field-expansion contribution (Section IV-D); deeper-stage blocks have progressively smaller and more uniform gate maps. Gate values are channel-averaged for visualisation.

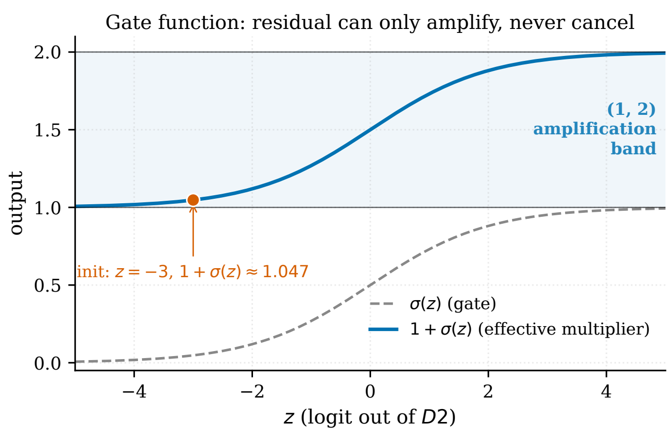

Fig. 3. Gate function $\sigma(z)$ (dashed) and effective multiplier $1+\sigma(z)$ (solid blue) over $z \in [-5, 5]$, with the $(1, 2)$ amplification band shaded. The D_2 initialisation point $z_{\text{init}} = -3$ gives $1 + \sigma(-3) \approx 1.05$, marked.

The gradient through a CBA block therefore cannot collapse, even at gate saturation — the multiplicative-gate analogue of He'16's gradient-highway property [6].

Proposition 3 (Bounded amplification)**.** *With* $\alpha = \sigma(z) \in (0, 1)^d$*, the effective per-coordinate multiplier* $1 + \alpha_i \in (1, 2)$*, so* $|X_i| < |Y_i| < 2|X_i|$.

Signal is amplified through the block, never suppressed. The gate function is illustrated in Figure 3.

The lineage in the literature converges on the same pattern — gate the residual term, never the identity [19], [24], [25]. CBA's per-pixel-per-channel choice generalises these (which gate per-channel or rank-1) to full $C \times H \times W$ granularity.

Initialisation. The terminal D_2 `project_bn.bias` is initialised to -3, giving $\sigma(-3) \approx 0.047$ and effective multiplier $1 + \alpha \approx 1.05$ at epoch 0 — block contributes $\sim$5% of identity at start and opens gradually (ReZero [25] warm-start principle). **DropPath** is applied per-sample to the gated residual only; the identity arm is never dropped. Linear schedule across the four CBA blocks: 0 at CBA-1 to `drop_path_rate` at CBA-4.

TABLE I
CBA BLOCK SPECIFICATIONS.

Block	Input	C_{mid}	C'	Latent	L	t
CBA-1	32×128^2	48	72	$16^2 \times 72$	2	4
CBA-2	96×32^2	112	128	$8^2 \times 128$	2	2
CBA-3	128×16^2	144	160	$8^2 \times 160$	3	2
CBA-4	160×8^2	144	128	$8^2 \times 128$	2	2

H. CBA block: formal description

The block is summarised as Algorithm 1. Input is $X \in \mathbb{R}^{B \times C \times H \times W}$; output is $Y = X + \text{DropPath}(\alpha \odot X)$ with $\alpha = \sigma(z)$ and z the output of the residual branch (encoder $\rightarrow$ tokenize $\rightarrow$ transformer $\rightarrow$ de-tokenize $\rightarrow$ decoder). Figure 4 traces the dataflow through one block; Figure 5 contrasts the resulting block against the MobileViT block it replaces.

V. ARCHITECTURE: CBA-MOBILEVIT-S

A. Backbone (unchanged from MobileViT-S)

The full network is shown in Figure 6. The backbone is byte-for-byte identical to MobileViT-S [8]: stem (Conv 3 $\rightarrow$ 16, 3×3, stride 2, BN, SiLU); Stage 1 (1× MV2 16 $\rightarrow$ 32, t=1, s=1); Stage 2 (3× MV2 32 $\rightarrow$ 64, t=4, first s=2); Stage 3 (1× MV2 64 $\rightarrow$ 96, t=4, s=2); Stage 4 (1× MV2 96 $\rightarrow$ 128, t=4, s=2); Stage 5 (1× MV2 128 $\rightarrow$ 160, t=4, s=2); head (1×1 Conv 160 $\rightarrow$ 640, BN, SiLU, AdaptiveAvgPool, FC 640 $\rightarrow$ 100). Feature map sizes at 256×256 input: 128 $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ 16 $\rightarrow$ 8.

B. CBA block instances

The four CBA blocks are sized for the $\sim$2.8 M-parameter budget along three axes: reduced bottleneck channels C'/C_{mid}, shallower deep stacks (CBA-3: L=3, CBA-4: L=2), and hybrid expansion ratio (t=4 at CBA-1, t=2 at CBA-2/3/4). Per-block specifications are in Table I.

Algorithm 1 CBA block forward pass (single block instance).

Require: Input feature map $X \in \mathbb{R}^{B\times C\times H\times W}$; encoder strides (s_1, s_2); channel widths C, C_{mid}, C'; expansion ratio t; transformer depth L, head count h, FFN factor $f{=}2$; DropPath rate $p \in [0, 1)$.
Ensure: Output feature map $Y \in \mathbb{R}^{B\times C\times H\times W}$ with $Y \approx X + (\sigma(z) \odot X)$.

Encoder, stage 1 (MV2-style, stride s_1):
  $z \leftarrow \mathrm{PW}_{C\to tC}(X)$; $z \leftarrow \mathrm{SiLU}(\mathrm{BN}(z))$ ▷ expand
  $z \leftarrow \mathrm{DW}_{3\times3,\,s_1}(z)$; $z \leftarrow \mathrm{SiLU}(\mathrm{BN}(z))$ ▷ spatial filter at expanded width
  $z \leftarrow \mathrm{PW}_{tC\to C_{\text{mid}}}(z)$; $z \leftarrow \mathrm{BN}(z)$ ▷ linear project (Sandler §3)
Encoder, stage 2 (MV2-style, stride s_2):
  $z \leftarrow \mathrm{PW} \circ \mathrm{DW}_{3\times3,\,s_2} \circ \mathrm{PW}_{\to C'}(z)$ ▷ $z \in \mathbb{R}^{B\times C'\times h'\times w'}$, $h' = H/(s_1 s_2)$

Tokenize: $z \leftarrow \mathrm{reshape}(z,\ B \times (h'w') \times C')$ ▷ spatial positions $\to$ tokens
Transformer (L **layers, pre-norm):**
for $\ell = 1$ **to** L **do**
  $z \leftarrow z + \mathrm{MHSA}_h(\mathrm{LN}(z))$ ▷ global mixing across $h'w'$ tokens
  $z \leftarrow z + \mathrm{FFN}_{f=2}(\mathrm{LN}(z))$ ▷ rank-collapse counterweight (Dong'21, Yu'24)
end for
De-tokenize: $z \leftarrow \mathrm{reshape}(z,\ B \times C' \times h' \times w')$

Decoder, stage 1 (MV2-style + bilinear $\times s_2$):
  $z \leftarrow \mathrm{PW}_{C'\to tC'}(z)$; $z \leftarrow \mathrm{SiLU}(\mathrm{BN}(z))$
  $z \leftarrow \mathrm{Bilinear}_{\times s_2}(z)$; $z \leftarrow \mathrm{DW} \circ \mathrm{PW}_{\to C_{\text{mid}}}(z)$; $z \leftarrow \mathrm{BN}(z)$
Decoder, stage 2 (MV2-style + bilinear $\times s_1$, terminal):
  $z \leftarrow \mathrm{PW} \circ \mathrm{Bilinear}_{\times s_1} \circ \mathrm{DW} \circ \mathrm{PW}_{\to C}(z)$
  $z \leftarrow \mathrm{BN}_{\beta=-3}(z)$ ▷ init: $\sigma(-3) \approx 0.05 \Rightarrow$ block $\approx$ identity

Gate and residual:
$\alpha \leftarrow \sigma(z) \in (0, 1)^{B\times C\times H\times W}$ ▷ per-pixel-per-channel gate
$r \leftarrow \alpha \odot X$ ▷ Hadamard amplification
if training **and** $p > 0$ **then**
  $r \leftarrow \mathrm{DropPath}_p(r)$ ▷ stochastic depth on gated branch only
end if
$Y \leftarrow X + r$ ▷ $+X$ identity arm preserves He'16 gradient highway
return Y

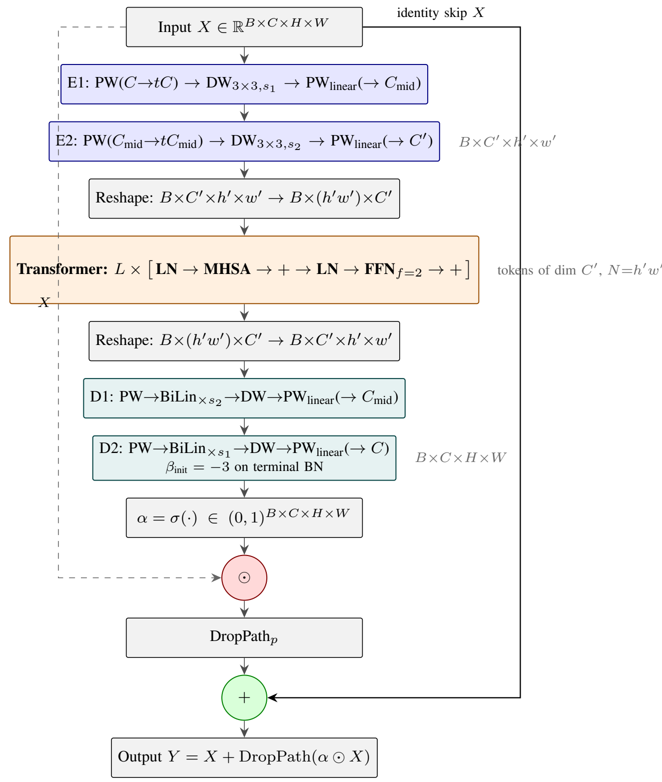

Fig. 4. CBA block internal architecture. The two-stage MV2-DSC encoder (E1, E2) compresses the input; the transformer reasons over the compressed latent; the symmetric two-stage decoder (D1, D2) maps the result back to the input shape. The terminal sigmoid produces the gate $\alpha \in (0, 1)^{B\times C\times H\times W}$, applied via Hadamard product inside an additive residual. The $+X$ arm preserves the He'16 gradient highway (Proposition 2).

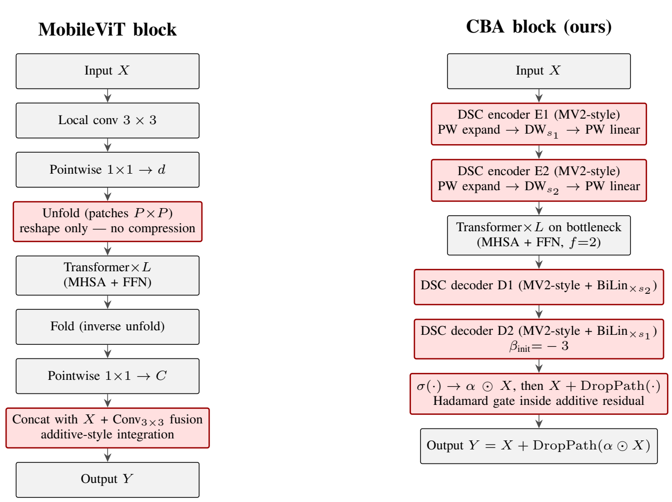

Fig. 5. Structural comparison of the MobileViT block (left) and the CBA block (right). Red-bordered boxes mark the differentiating components: MobileViT uses unfold/fold reshape (no spatial compression) followed by concat-fusion; CBA uses a two-stage non-linear DSC encoder/decoder around attention with Hadamard-gated additive residual integration.

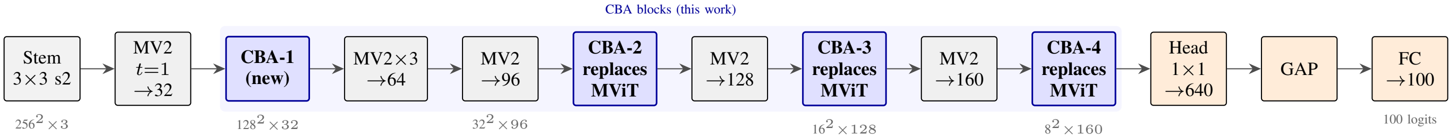

Fig. 6. CBA-MobileViT-S network skeleton. The convolutional backbone (stem + MV2 stages) is byte-for-byte identical to MobileViT-S [8]; the four CBA blocks (highlighted in blue) replace MobileViT-S's three transformer blocks at stages 3/4/5 and add one new upstream block (CBA-1) after stage 1. CBA-1's placement at 128×128 spatial extent is the load-bearing instance of the early-receptive-field contribution (Section IV-D).

TABLE II
AUTHORITATIVE PER-BLOCK MAC BREAKDOWN (MEASURED).

Block	MACs (M)	Fraction
Stem (Conv 3 → 16, s=2)	7.6	0.7%
Stage 1 (MV2 16 → 32)	12.3	1.1%
CBA-1	**297.5**	**26.7%**
Stage 2 (MV2 ×3, → 64)	407.9	36.6%
Stage 3 (MV2 → 96)	97.5	8.8%
CBA-2	93.6	8.4%
Stage 4 (MV2 → 128)	52.3	4.7%
CBA-3	76.4	6.9%
Stage 5 (MV2 → 160)	22.7	2.0%
CBA-4	39.3	3.5%
Head 1×1 + AvgPool	6.7	0.6%
FC 640 → 100	0.1	0.0%
Total	**1113.7**	**100.0%**

C. Parameter and compute breakdown

The trained model has 2,785,956 parameters and 1.114 G MACs (`fvcore`) at 256×256; Table II gives the per-block breakdown. Stage 2 (unchanged from MobileViT-S) contributes the largest single fraction at 36.6% but is shared with the baseline. The four CBA blocks contribute 45.5% together, dominated by **CBA-1 at** 26.7% (about $6\times$ the other three combined): CBA-1 sits at the highest spatial extent and runs a bilinear $\times4$ upsample at its terminal D_2 stage, so the cost concentrates where the receptive-field-expansion contribution does.

VI. EXPERIMENTAL SETUP

The recipe is the DeiT / MobileViTv2-family standard at 50-epoch length on ImageNet-100. Per-knob comparison across the three runs is in Table III.

Optimizer. AdamW [26], decoupled weight decay 0.05 (ndim$\geq$ 2 only), 10-epoch linear warmup then cosine to 0, gradient-clip at norm 1.0, label smoothing 0.1.

Data and augmentation. ImageNet-100 ($\sim$126.7 K train / 5 K val) at 256×256. RandomResizedCrop, horizontal flip, RandAugment $(N{=}2, M{=}9)$ [27], per-batch Mixup (α=0.2) [28] or CutMix (α=1.0) [29] with p=0.5.

Regularisation. Stochastic depth [30] on the gated branch only, linear $0 \to 0.05$ across CBA-1 . . . CBA-4. Model EMA decay 0.995; checkpoint is $\max(\text{live}, \text{EMA})$.

Hardware. CBA-lite and MobileViT-XS: NVIDIA RTX PRO 6000 Blackwell, bf16 autocast, $\sim$3–4.5 h per run. MobileViTv2-0.75: Kaggle T4, fp16 + GradScaler, $\sim$10 h.

VII. RESULTS

CBA-MobileViT-S-lite reaches 81.52% **top-1 (EMA)** on ImageNet-100 at 50 epochs — $+4.62$ pp over the matched-protocol MobileViT-XS control at $+0.18$ G MACs, and $+3.48$ pp over the asymmetric MobileViTv2-0.75 reference at $+0.06$ G MACs. All runs trained from scratch at 256×256. CBA-lite and MobileViT-XS are byte-for-byte parity-locked (Blackwell, B=512, peak LR 1.5×10^{-3}, decoupled-WD AdamW, 10-epoch warmup + cosine, RandAug + Mixup/CutMix, drop_path 0.05, EMA decay 0.995, seed 42); the v2-0.75 run used a different schedule on a Kaggle T4 (disclosed below).

A. Headline results

The MAC overheads are $+19\%$ over MobileViT-XS and $+6\%$ over MobileViTv2-0.75 — both small relative to the accuracy gaps. The accuracy-vs-MACs Pareto plot is Figure 8.

On the v2-0.75 control. The MobileViTv2-0.75 run used batch size 128 with peak LR 4×10^{-4} on a Kaggle T4 — $\sim$4$\times$ more optimizer steps per epoch and a marginally higher LR-to-batch ratio than CBA-lite. Both effects favour v2-0.75 under a fixed 50-epoch schedule, so the +3.48 pp CBA-lite advantage is understated, not overstated.

On deployment scope. The accuracy / parameter / theoretical-FLOPs comparison is the strongest claim the available compute supports. Wall-clock latency on mobile CPU, NPU, or MCU is not measured — edge-deployment testing requires runtime export, per-device tuning, and hardware access beyond the project budget. The headline is a *theoretical* efficiency result; deployment evaluation is deferred (Section VIII-D).

B. Training trajectory

Figure 7 plots per-epoch curves for all three runs. The CBA-lite variant trains cleanly: no warmup-spike events, monotonic loss decrease through epoch 10, and a val/EMA gap that stays within the expected 0–4 pp range. The EMA model tracks live within ±0.2 pp by mid-training.

VIII. DISCUSSION

A. What the result establishes

CBA's claim is that **deep bottleneck attention belongs at high spatial extent**. The 81.52% top-1 (+4.62 pp over the matched MobileViT-XS) is the feasibility result; the architectural payoff is full receptive field at a stage where pure

TABLE III
TRAINING-SETUP COMPARISON ACROSS THE THREE LITE-BUDGET RUNS. GREEN = MATCHED ACROSS ALL THREE; YELLOW = MATCHED BETWEEN CBA-LITE AND MOBILEVIT-XS ONLY (THE DISCLOSED MOBILEVITV2-0.75 ASYMMETRY).

Training-setup comparison — matched protocol vs. disclosed asymmetries

Setting	CBA-MobileViT-S-lite (ours)	MobileViT-XS	MobileViTv2-0.75	Status
Hardware (GPU)	RTX PRO 6000 Blackwell	RTX PRO 6000 Blackwell	Kaggle (Tesla-class)	Matched (CBA↔XS); v2-0.75 differs
Resolution	256×256	256×256	256×256	Matched (all three)
Dataset	ImageNet-100	ImageNet-100	ImageNet-100	Matched (all three)
Epochs	50	50	50	Matched (all three)
Batch size	512	512	128	Matched (CBA↔XS); v2-0.75 differs
Steps / epoch	247	247	~990	Matched (CBA↔XS); v2-0.75 differs
Peak LR	1.5e-3	1.5e-3	4.0e-4	Matched (CBA↔XS); v2-0.75 differs
LR / batch ratio	2.93e-6	2.93e-6	3.13e-6	Matched (CBA↔XS); v2-0.75 differs
Warmup epochs	10	10	5	Matched (CBA↔XS); v2-0.75 differs
LR schedule	cosine to 0	cosine to 0	cosine to 0	Matched (all three)
Optimizer	AdamW	AdamW	AdamW	Matched (all three)
Weight decay (decoupled)	0.05	0.05	0.05	Matched (all three)
Gradient clip (norm)	1.0	1.0	1.0	Matched (all three)
Label smoothing	0.1	0.1	0.1	Matched (all three)
RandAugment (N, M)	(2, 9)	(2, 9)	(2, 9)	Matched (all three)
Mixup α	0.2	0.2	0.2	Matched (all three)
CutMix α	1.0	1.0	1.0	Matched (all three)
Mixup/CutMix prob	0.5 (50/50 split)	0.5 (50/50 split)	0.5 (50/50 split)	Matched (all three)
Stochastic depth	0.05	0.05	0.05	Matched (all three)
EMA decay	0.995	0.995	0.999	Matched (CBA↔XS); v2-0.75 differs
Mixed precision	bf16	bf16	bf16	Matched (all three)
Random seed	42	42	42	Matched (all three)
Dataloader workers	8	8	8	Matched (all three)

Matched (all three) · Matched (CBA↔XS); v2-0.75 differs · Differs (all three)

TABLE IV
LITE-BUDGET EVALUATION ON IMAGENET-100, 50 EPOCHS, 256×256. *Best top-1* IS $\max$(LIVE, EMA).

Model	Params	MACs (fvcore)	Val top-1	Val top-5	EMA top-1	Best top-1
CBA-MobileViT-S-lite (ours)	**2.79 M**	**1.114 G**	81.32%	95.76%	**81.52%**	**81.52%**
MobileViT-XS [8]	1.97 M	0.935 G	76.78%	94.52%	76.78%	76.90%
MobileViTv2-0.75 [9]	2.52 M	1.051 G	77.90%	94.40%	77.86%	78.04%

CNN delivers ∼11-pixel local features. The Hadamard-gated additive residual is a supporting choice, not the contribution.

B. Where the architecture should and shouldn't help

CBA should help where evidence from distant spatial positions matters early — fine-grained recognition, scene-conditioned classification, occlusion-heavy benchmarks, and corruption robustness — and Figure 2 shows compute concentrating on object regions consistent with this. It should *not* help where local features are already sufficient (coarse texture, centred objects, micro-images), and the budget is fixed: CBA-1 alone consumes 26.7% of total MACs, so dynamic-token-pruning [16], [18] retains a flexibility advantage there.

C. Limitations

- **ImageNet-100 only.** Generalisation to IN-1K, longer schedules, and downstream tasks (segmentation, detection) is future work.
- **Single-seed numbers.** Single-seed variance at this scale is typically 0.3–0.7 pp; the +4.62 pp / +3.48 pp gaps sit comfortably outside that band, but a 3-seed bound is the publishable evidence.
- **Single-budget evaluation.** A larger-budget variant and its matched MobileViTv2-1.0 control are deferred to a compute-funded extension.
- **No 2024-class hybrid baseline.** EdgeNeXt [31], FastViT [11], EfficientFormerV2 [10], RepViT [12] are not in the headline.
- **No edge-device deployment.** The efficiency claim is *theoretical*; wall-clock latency is out of scope.
- **No dense-prediction transfer.** Segmentation (e.g., ADE20K [32]) and detection are future work.

D. Future work

The highest-priority follow-ups: (1) multi-seed re-runs of the headline (∼9 GPU-hours, 3 seeds); (2) the *CBA-1 ablation* (drop the early block, keep CBA-2/3/4) — the cleanest test

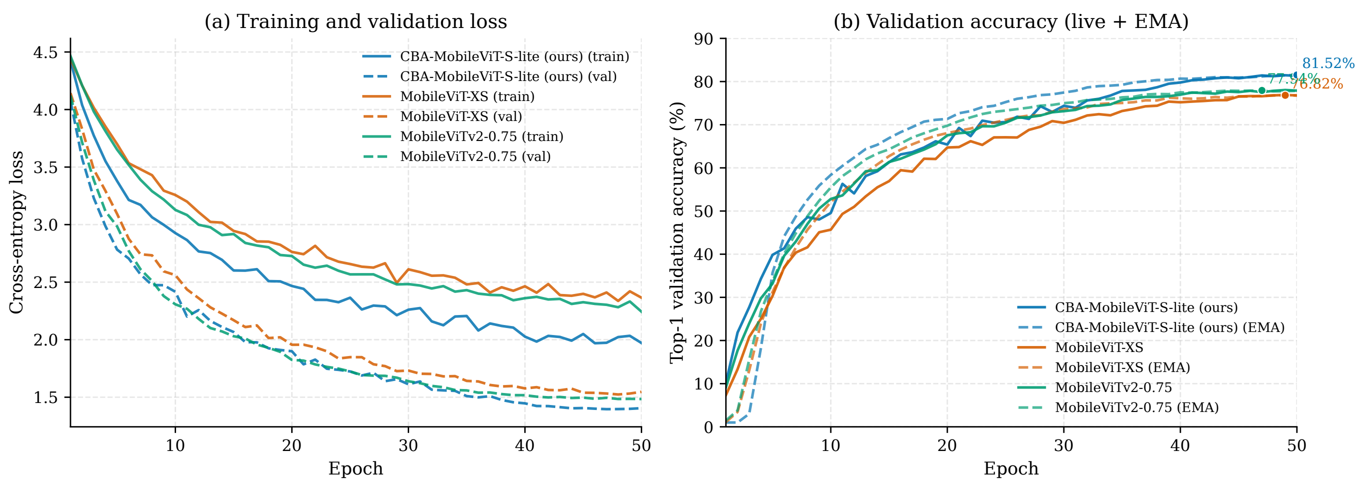

Fig. 7. Training trajectory across 50 epochs. Left: train loss (solid) and validation loss (dashed) for all three lite-budget runs. Right: top-1 validation accuracy (live) and EMA (dashed); best EMA points annotated.

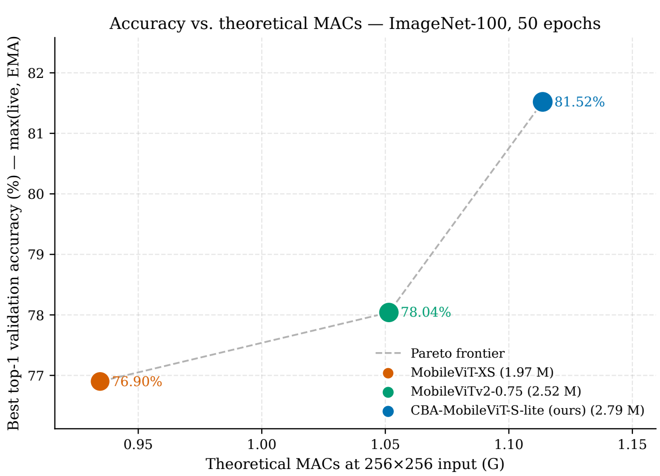

Fig. 8. Accuracy vs. theoretical MACs. ImageNet-100, 50 epochs. y-axis: best top-1 = $\max$(live, EMA). Point area proportional to parameter count.

of the early-receptive-field claim of Section IV-D; (3) the inverse *CBA-only-1 ablation* (keep only the early block); (4) a latent-geometry probe on CBA-1's encoder output (within- vs. between-class cosine similarity and linear-probe accuracy), converting the open question of Section IV-B into measured data; (5) `noFFN` and `additive` ablations testing Sections IV-F and IV-G. Beyond ablations: edge-device deployment (TFLite / Core ML / ONNX-Runtime), a larger-budget variant against MobileViTv2-1.0, comparison against a 2024-class hybrid baseline (e.g., EdgeNeXt-XS [31]), knowledge distillation, and downstream transfer to segmentation (ADE20K [32]) and detection (COCO [33]).

IX. Conclusion

Hybrid mobile vision transformers defer attention to the late stages of the network, leaving early stages pure-convolutional with locally-bounded receptive field. **Deep bottleneck attention** is the natural alternative; the structural concerns that left it unoccupied — latent geometry and compute cost — are addressed by treating the first as an open empirical question grounded in end-to-end self-consistency, and the second by convolutional operators in the MobileNetV2-style depthwise-separable variant. The architectural payoff is full receptive field at the 128×128 output of MobileViT-S's stage 1, where standard hybrids still operate on $\sim$11-pixel local patches. CBA-MobileViT-S-lite (2.79 M params, 1.114 G MACs) trained on ImageNet-100 reaches 81.52% **top-1 (EMA)** — $+$**4.62 pp** over the matched MobileViT-XS control and $+$**3.48 pp** over MobileViTv2-0.75. The 2024-frontier positioning, multi-seed bound, and edge-deployment evaluation are extensions catalogued in Section VIII-D.

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